
Proof Methods

Four ways to argue that something is true, and one way to show it isn't.

Today

- 01 Theorem, lemma, corollary, conjecture
- 02 Direct proof
- 03 Proof by contraposition
- 04 Proof by contradiction
- 05 Disproof by counterexample

direct proof



proof by
contradiction



proof is left as
an exercise
to the reader



proof is trivial



proof by lack of
counterexample



The Vocabulary of Proof

TERMINOLOGY

Theorem	A statement that can be shown to be true, and that someone thought was worth naming.
Proposition	A theorem of lesser importance. Same standard of proof, less fanfare.
Proof	A valid argument that establishes the truth of a theorem.
Axiom	A statement we assume to be true. Everything else is built on these.
Lemma	A smaller result proved because it makes a bigger proof possible.
Corollary	A theorem that follows directly from one already proved.
Conjecture	A statement proposed as true on partial evidence or expert intuition. Not proved.

What Counts as a Proof

TERMINOLOGY

A PROOF IS

A finite chain of statements.

Each one follows from the assumptions, an axiom, or a result already established.

The last one is the theorem.

A PROOF IS NOT

A list of examples.

A restatement of the claim in different words.

The word “clearly” doing the load-bearing work.

Direct Proof

METHOD 1

$$p \rightarrow q$$

Assume p . Use definitions and known results to reach q .

You never touch the rows where p is false. Those rows are true for free.

THE SHAPE

1. Suppose p holds.
2. By definition, p means ...
3. Therefore ...
4. Which is exactly q . ■

Example: Odd n Gives Odd n^2

METHOD 1

THEOREM

If n is an odd integer, then n^2 is odd.

Proof. Assume n is odd. By definition, $n = 2k + 1$ for some integer k .

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

Since $2k^2 + 2k$ is an integer, n^2 is one more than twice an integer — which is the definition of odd. ■

The whole proof is one algebraic regroup. The skill is knowing which form to aim for.

Proof by Contraposition

METHOD 2

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Prove the contrapositive instead. It is the same statement, so proving it proves the original.

Reach for it when $\neg q$ hands you something concrete and p does not.

THE SHAPE

1. Suppose q is false.
2. Work forward from $\neg q$.
3. Conclude p is false.
4. So $p \rightarrow q$ holds. ■

Lemma: If n^2 Is Even, So Is n

METHOD 2

LEMMA

For every integer n : if n^2 is even, then n is even.

Proof. We prove the contrapositive: if n is odd, then n^2 is odd. That is the theorem we just proved. ■

Direct proof would start from “ n^2 is even,” which gives you $n^2 = 2m$ and nowhere obvious to go. The contrapositive hands you $n = 2k + 1$ on line one.

Keep this one. We use it twice before the end of class.

Example: $n = ab$ Forces a Small Factor

METHOD 2

THEOREM

If $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Proof. Suppose the conclusion fails: $\neg ((a \leq \sqrt{n}) \vee (b \leq \sqrt{n}))$.

By De Morgan, that means $a > \sqrt{n}$ *and* $b > \sqrt{n}$. Both sides are positive, so we may multiply the inequalities:

$$ab > \sqrt{n} \cdot \sqrt{n} = n$$

So $ab \neq n$ — the hypothesis is false. Having shown $\neg q \rightarrow \neg p$, we have shown $p \rightarrow q$. ■

This is contraposition, not contradiction. We never assumed the hypothesis — we concluded it was false. That distinction is worth points.

Proof by Contradiction

METHOD 3

assume $\neg s$ derive F

To prove a statement s , assume it is false and derive something impossible. Since $\neg s$ led to an absurdity, $\neg s$ is false, so s is true.

For a conditional $p \rightarrow q$, the negation is $p \wedge \neg q$ — so you assume both.

THE SHAPE

1. Suppose, for contradiction, $\neg s$.
2. Reason forward.
3. Reach $r \wedge \neg r$.
4. Therefore s . ■

METHOD 3, EMOTIONALLY

“Assume I’m wrong. ...that went badly. So I’m right.”

It feels like a trick the first ten times. It is just $\neg(\neg s) \equiv s$ wearing a disguise.



Rational Numbers

METHOD 3

DEFINITION

A real number x is **rational** if there exist integers p and q with $q \neq 0$ such that $x = p/q$. A real number that is not rational is **irrational**.



The set of rationals.

Every rational can be written in lowest terms — no common factors. We lean on that in twenty minutes.

Example: The Sum of Two Rationals

METHOD 1

THEOREM

The sum of two rational numbers is rational.

Proof. Let r and s be rational. Then $r = p/q$ and $s = t/u$ for integers p, q, t, u with $q \neq 0$ and $u \neq 0$.

$$r + s = p/q + t/u = (up + tq) / uq$$

Both $up + tq$ and uq are integers, and since $q \neq 0$ and $u \neq 0$, we have $uq \neq 0$. So $r + s$ is a ratio of integers with nonzero denominator, hence rational. ■

Checking $uq \neq 0$ is not pedantry. Without it you have not met the definition.

Example: $\sqrt{2}$ Is Irrational

METHOD 3 — PART I

Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational.

Then there are integers a and b with $b \neq 0$ and $\sqrt{2} = a/b$, where a and b have **no common factors** — every rational can be written in lowest terms.

$$\sqrt{2} = a/b \implies 2 = a^2/b^2 \implies 2b^2 = a^2$$

So a^2 is even. By our lemma, a is even, so $a = 2c$ for some integer c .

Hold on to “no common factors.” Everything hinges on it.

Example: $\sqrt{2}$ Is Irrational

METHOD 3 — PART II

$$a = 2c \implies a^2 = 4c^2 \implies 2b^2 = 4c^2 \implies b^2 = 2c^2$$

So b^2 is even. By the lemma again, b is even.

Now a and b are both even, so 2 divides both — **contradicting** the assumption that they have no common factors.

The assumption that $\sqrt{2}$ is rational is therefore false. $\sqrt{2}$ is irrational. ■

FOOTNOTE

Roughly 2,500 years old, and legend has it the Pythagoreans were not pleased. The argument has not needed a patch since.

Disproof by Counterexample

METHOD 4

$$\neg (\forall x; P(x)) \equiv \exists x; \neg P(x)$$

To refute “every x has property P ,” exhibit one x that doesn’t. Then stop.

The reverse never works: no pile of examples proves a universal claim.

Example: Sums of Two Squares

CLAIM (FALSE)

Every positive integer is the sum of the squares of two integers.

Proof. Take $n = 3$. The only perfect squares not exceeding 3 are $0^2 = 0$ and $1^2 = 1$. Every sum of two of them is listed at right, and none of them is 3. ■

Pair	Sum
$0^2 + 0^2$	0
$0^2 + 1^2$	1
$1^2 + 1^2$	2
nothing	3

Choosing a Method

PRACTICE

Direct	The hypothesis gives you an equation or a structure to work with.
Contraposition	The negated conclusion is concrete and the hypothesis is not. Often when the conclusion is an “or.”
Contradiction	The claim is a negative — nothing exists, no such number, cannot be written that way.
Counterexample	You suspect the claim is false. Try small cases before you try to prove anything.

Try direct first, always. If it stalls within two lines, write down $\neg q$ and see what it hands you.

Common Mistakes

BEFORE YOU GO

- 01 Proving a universal claim with examples. Three cases that work prove nothing about the fourth.
- 02 Assuming the conclusion. If the thing you are proving appears as a step, the proof is circular.
- 03 Confusing contraposition with contradiction. Contraposition assumes $\neg q$ only; contradiction assumes $p \wedge \neg q$.
- 04 Using a definition without writing it down. “Odd” means $n = 2k + 1$. Put that line in the proof.

Recap

LECTURE 03

- 01 A proof is a chain of justified statements ending at the theorem.
- 02 Direct: assume p , walk to q . Start by unfolding the definition.
- 03 Contraposition: prove $\neg q \rightarrow \neg p$. Legal because the two are equivalent.
- 04 Contradiction: assume the negation, derive an absurdity.
Best (but not always!) for negative claims.
- 05 One counterexample refutes a universal claim.
No number of examples proves one.



Next: set theory.